3 Figure 3.1 shows an electric shower.

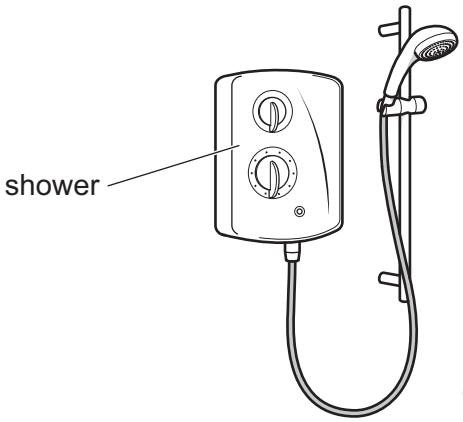


Figure 3.1

The temperature of water as it enters the shower is 15 °C. The water is heated in the shower. The water leaves the shower at a temperature of 38 °C.

(a) The density of water is 1000 kg / m^3.
The specific heat capacity of water is 4200 J / (kg °C).

(i) The water flows through the shower at a rate of 0.0045 m^3 / min.

Show that the mass of water that leaves the shower in 4.0 min is 18 kg.

[2]

(ii) Calculate the energy required to increase the temperature of 18 kg of water from 15 °C to 38 °C.

energy = ... J [2]

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(iii) The energy in **3(a)(ii)** is transferred to the water in 4.0 min.

Calculate the power transferred to the water.

power = ... W [2]

(b) At another time, the temperature of the water leaving the shower is higher. The power supplied to the water remains constant.

Suggest **one** reason for this increase in temperature.

.....

..... [1]

[Total: 7]

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